

Summary: M.S. in Biomedical Engineering, ShanghaiTech University. Experienced in non-Cartesian pulse sequence design and reconstruction, with publications in IEEE TBME / ISMRM / SCMR / AAAI / JMRI, and open-source projects on PyPI / GitHub, invited as reviewer by IEEE TMI / ISMRM.

Skills: Pulse Sequence Design, Reconstruction, Artifacts Correction, Simulation, Python (with C-API, CUDA acceleration), MATLAB (with MEX interface), C/C++, CUDA, Linux, Circuit Design, Embedded Programming, Desktop Development (Qt)

Education

- 2023 - 2026 **M.S. (Biomedical Engineering) at ShanghaiTech University** (major GPA: 3.92/4.00)
(overall GPA: 3.79/4.00)
Major Courses: Algorithms (**A+**), Principles of MRI (**A**), Medical Image Processing (**A**), MRI System Design (**A**), Semiconductor Detectors and Readout Electronics (**A**)
- 2018 - 2022 **B.S. (Automation) at Shandong University** (overall GPA: 81.31/100)
Awards: 1× National **1st** Prize, 1× National **2nd** Prize, 1× Provincial **1st** Prize.

Publications

- [1] 2026, Real-Time Gradient Waveform Design for Arbitrary k-Space Trajectories **Pulse Sequence**
- ▷ **Target:** Develop a general algorithm for calculating non-Cartesian gradient waveforms.
 - ▷ **Significance:** Improve the computational efficiency by **1 order of magnitude**, achieving real-time gradient design. Improve the slew-rate accuracy by **2 orders of magnitude**.
 - ▷ **Role:** **1st author**. Designed and implemented the proposed algorithm, and a trajectory library built on it.
 - ▷ **Status:** Published at <https://doi.org/10.1109/TBME.2026.3654117> on *IEEE Trans. Biomed. Eng.*
- [2] 2026, Sampling Density Compensation using Fast Fourier Deconvolution **Reconstruction**
- ▷ **Target:** Develop a general algorithm for calculating density compensation functions.
 - ▷ **Significance:** Improve the computational efficiency by **1-2 orders of magnitude**, saving tens of minutes.
 - ▷ **Role:** **1st author**. Designed and implemented the proposed algorithm.
 - ▷ **Status:** Published at <http://echo.ismr.org/p/ISMRM2026/461-03-015> on *ISMRM-2026*.
- [3] 2026, Zero-shot INMR for Ultra-high Temporal Resolution Dynamic MRI
- ▷ **Role:** **Coauthor**. Assistant in data acquisition using the algorithm in [1].
 - ▷ **Status:** Published at <https://doi.org/10.1609/aaai.v40i5.37395> on *AAAI-2026*.
- [4] 2026, Detection of Coronary Microvascular Dysfunction in Diabetic Mice Using Arterial Spin Labeling Cardiac MRI: A Multimodality Imaging Comparison
- ▷ **Role:** **Coauthor**. Maintained an arterial perfusion marker (ASL) sequence on a Biospec 94/30 scanner.
 - ▷ **Status:** Published at <https://doi.org/10.1002/jmri.70304> on *J. Magn. Reson. Imaging*.
- [5] 2025, ASL-CMR detects CMD in diabetic mice: a multi-imaging modality comparison
- ▷ **Role:** **Coauthor**. Maintained an arterial perfusion marker (ASL) sequence on a Biospec 94/30 scanner.
 - ▷ **Status:** Published at <https://archive.ismr.org/2025/1028.html> on *ISMRM-2025*.

Open-Source Projects

MRArbGrad **Pulse Sequence**

- ▷ **Intro.:** The fastest non-Cartesian gradient solver at present; **Real-time capability:** **gradients solve time is shorter than its duration**; Written in C++ backend with a Python wrapper; Compatible with all pulse se-

quence development platforms using C++ or Python; Includes a ready-to-use trajectory library; Manuscript Reference: [1].

- ▷ Link (PyPI): <https://pypi.org/project/MRArbGrad>
- ▷ Link (GitHub): <https://github.com/rui-luo1002/MRArbGrad>
- ▷ Role: **Sole Developer**.

MRArbDcf Reconstruction

- ▷ Intro.: The fastest density compensation function (DCF) solver at present; **DCF solve time: 0.1 sec for 2D or 20 sec for 3D (CPU)**; Supports both CPU and GPU platforms. Manuscript Reference: [2].
- ▷ Link (PyPI): <https://pypi.org/project/MRArbDcf>
- ▷ Link (GitHub): <https://github.com/rui-luo1002/MRArbDcf>
- ▷ Role: **Sole Developer**.

TorchFinufft-Ryan Optimization

- ▷ Intro.: NUFFT module for PyTorch and a Toeplitz MSE loss module; Useful in optimization-based or deep-learning-based reconstruction projects; **In 256^2 iterative reconstruction: 2 ms per step using NUFFT or 3 ms per step using Toeplitz on an RTX3090.**
- ▷ Link (PyPI): <https://pypi.org/project/torchfinufft-ryan>
- ▷ Link (GitHub): <https://github.com/rui-luo1002/torchfinufft>
- ▷ Role: **Sole Developer**.

MRPhantom Simulation

- ▷ Intro.: Volumetric dynamic MRI Phantom of a Slime with respiratory and cardiac motion, and M0, phase, T1, T2, B0 maps, and also coil sensitivity maps; Boosted by a parallel C-API Backend; **Support memory-efficient mode: phantoms are generated in real time according to motion parameter to avoid explicit 3D+T construction (137 ms per 256^3 volume).**
- ▷ Link (PyPI): <https://pypi.org/project/MRPhantom>
- ▷ Link (GitHub): <https://github.com/rui-luo1002/MRPhantom>
- ▷ Role: **Sole Developer**.

Awards

National 1st prize in Smart Car Competition, Acoustic Group, 2020 Signal Processing

- ▷ Target: Design a model car to hit certain beacons playing acoustic chirp signals.
- ▷ Selectivity: 1 team per category per school.
- ▷ Role: **Team member**. Developed the fast cross-correlation algorithm for acoustic navigation on a micro-controller, and also a secondary optical navigation algorithm.

National 2nd prize in Smart Car Competition, Delivery Group, 2021 Circuit Design

- ▷ Target: Design a model car with LiDAR and Linux microcomputer to deliver packages in a maze.
- ▷ Selectivity: 1 team per category per school.
- ▷ Role: **Team leader**. Full-stack experience including circuit design and assembly, robot operating system debugging, optical navigation and PID algorithm development, TCP host application development, hardware driver development.

Provincial 1st prize in Smart Car Competition, Optical Group, 2019 Embedded Programming

- ▷ Target: Design a model car to trace the racing track using a digital camera.
- ▷ Selectivity: 2 teams per category per school.
- ▷ Role: **Team member**. Developed the optical navigation algorithm using Otsu's threshold algorithm, edge detection algorithm and PID control algorithm.

Other Awards: Two additional national prizes and three university-level prizes for which I was invited to provide technical support.

Experience

2025 September, Reviewer, *IEEE Transactions on Medical Imaging*

- ▷ **Role:** **Reviewer.**
- ▷ **Highlights:** Reviewed one manuscript.

2025 November, Reviewer, *International Society of Magnetic Resonance in Medicine*

- ▷ **Role:** **Reviewer.**
- ▷ **Highlights:** Reviewed 20 abstracts.

2024 Fall, Teaching Assistant, *Principles of Magnetic Resonance Imaging*

- ▷ **Role:** **Teaching Assistant.**
- ▷ **Highlights:** Mentored 33 students; designed the final project involving sampling and reconstruction simulation; delivered a review lecture highly rated by students.

Reference

